	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Control Systems	3	0	0	3	3
	PC	Control Systems Lab	0	0	2	2	1
	PC	Linear Integrated Circuits and Applications	3	0	0	3	3
	PC	Linear Integrated Circuits and Applications Laboratory	0	0	2	2	1
	PC	Digital Communication	3	0	0	3	3
	PC	Digital Communication Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VI	OE	Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Computer and Communication Networks	3	0	0	3	3
	PC	Computer and Communication Networks Laboratory	0	0	2	2	1
	PC	Computer Organization and Microprocessor	3	0	0	3	3
	PC	Computer Organization and Microprocessor Lab	0	0	2	2	1
	PC	Microwave and Antenna	3	0	0	3	3
	PC	Microwave and Antenna Lab	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 4 (3-0-0)	3	0	0	3	3
	PE	Program Elective 5 (3-0-0)	3	0	0	3	3
	PC	Modern Wireless Communication	3	0	0	3	3
	PC	Modern Wireless Communication Lab	0	0	2	2	1
	PC	Embedded System Design	3	0	0	3	3
	PC	Embedded System Design Lab	0	0	2	2	1
	PC	Digital CMOS VLSI Design	3	0	0	3	3
	PC	Digital CMOS VLSI Design Lab	0	0	2	2	1
		Seminar					1
			18	**0**	**6**	**24**	**24**
Sem VIII		**Course Name**					
	Pro	Major/Comprehensive Project					**12**

Program Core Electives:

Semester	VLSI/Embedded systems	Communication Systems	Computer systems	Control
5	Digital Systems Design using HDL	Opto Electronics and Optical Communication		
6	Analog IC Design	Satellite Communication	Machine Learning and Applications	Introduction to Robotics
	Power Electronics	Information Theory and Coding	Database Management Systems	Modern Control Systems
7	Advanced Processors and SoCs	Advanced Communication Networks	Internet of Things	Drones: Design, Theory and Applications
	IC Technology	Radar and Navigation Systems	Deep Learning and Applications	
	Mixed Signal VLSI Design			

Semester 7

24ICxxxT					Modern Wireless Communications					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Understand wireless communication evolution, cellular concepts, and system design.
- Analyze radio wave propagation and multipath fading in wireless systems.
- Evaluate performance of wireless communication systems in the presence of fading.
- Explore diversity, multiplexing, and access techniques for enhanced wireless communication.

UNIT 1: Introduction to Wireless Communications	**09 Hrs.**
Evaluation of Wireless Communications (2G to 5G); Cellular Concept: System Design Fundamentals, Frequency Reuse, Handoff, Cochannel Interference and System Capacity	
UNIT 2: Mobile Radio Propagation	**12 Hrs.**
Path Loss and Shadowing: Radio Wave Propagation, Transmit & Receive Signal Models, Free-Space Path Loss, Shadow Fading; Statistical Multipath Channel Models: Narrowband and Wideband Fading, Rayleigh and Rician Fading Distribution; Channel Equalizer; Performance Analysis over Fading Channel: Outage Probability, Average Probability of Error, Capacity	
UNIT 3: Space-Time Diversity and Multiplexing	**11 Hrs.**
SIMO, MISO and MIMO Models; Receiver Diversity: Selection Combining, Equal Gain Combining, Maximal-Ratio Combining, Transmit Diversity: Alamouti Space-Time Coding Scheme. Spatial Multiplexing.	
UNIT 4: Multiple Access Techniques	**10 Hrs.**
Spread Spectrum Communication: Spread Spectrum Principles, Direct Sequence Spread Spectrum, Frequency Hoping Spread Spectrum; Multicarrier Modulation Schemes: Orthogonal Frequency Division Multiplexing (OFDM), Cyclic Prefix, Case Study: The IEEE 802.11a Wireless LAN Standard; Introduction to Next Generation Wireless Communication Technologies.	

Total: 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Comprehend wireless system evolution, cellular concepts, and design fundamentals.

CO2: Analyze radio wave propagation, path loss, fading models, and multipath effects.

CO3: Evaluate wireless communication system performance under fading conditions.

CO4: Apply diversity, multiplexing, and access techniques for enhanced communication.

CO5: Understand multiple antenna systems to improve communication performance.

CO6: Utilize various multiple access techniques in modern wireless communication systems.

TEXT/REFERENCE BOOKS

1. A. Goldsmith, “Wireless Communications,” Cambridge University Press, 2005.
2. D. Tse and P. Viswanath, “Fundamentals of Wireless Communications,” Cambridge University Press, 2005.
3. T. S. Rappaport, “Wireless Communications: Principles and Practice,” 2nd Ed., Prentice Hall, 2002.
4. Yifei Yuan, Zhifeng Yuan, “5G New Radio Non-Orthogonal Multiple Access”, CRC Press, 2023.

<Course Code>					Modern Wireless Communication Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	0	0	0	50	50	100

COURSE OBJECTIVES:

- Understand wireless communication evolution, cellular concepts, and system design.
- Analyze radio wave propagation and multipath fading in wireless systems.
- Evaluate performance of wireless communication systems in the presence of fading.
- Explore diversity, multiplexing, and access techniques for enhanced wireless communication.

List of Experiments:

Following is the list of experiments of experiments, but it is not limited to it.

1. Generating PDF and CDF of diverse random variables and validating with Histogram analysis.
2. Simulating and analyzing path loss and shadowing models in wireless communication systems.
3. Investigating the impact of AWGN noise on data transmission quality.
4. Simulating and exploring different fading channel models in wireless communication scenarios.
5. Analyzing the outage probability for a SISO wireless communication system through simulation.
6. Evaluating the ASER of wireless communication systems under Rayleigh fading conditions.
7. Implementing the OFDM scheme with integrated channel equalization for performance evaluation.
8. Investigating and contrasting spatial diversity techniques such as SISO, MISO, SIMO, and MIMO.
9. Generating BER plots for different spatial diversity techniques to analyze their performance.
10. Studying various combining techniques including MRC, EGC, and Selection Combining for wireless systems.
11. Implementing and analyzing the Alamouti space-time coding scheme.

COURSE OUTCOMES:

On completion of the course, student will be able to:

CO1: Understand the principles of modern wireless communication systems.
CO2: Analyze and model the propagation characteristics of wireless channels.
CO3: Design and evaluate diversity techniques to enhance wireless communication reliability.
CO4: Implement advanced modulation and coding schemes for efficient data transmission.
CO5: Investigate the impact of noise and fading on wireless communication system performance.
CO6: Develop and optimize wireless systems for maximum capacity and coverage.

TEXT/REFERENCE BOOKS

1. A. Goldsmith, “Wireless Communications,” Cambridge University Press, 2005.
2. D. Tse and P. Viswanath, “Fundamentals of Wireless Communications,” Cambridge University Press, 2005.
3. T. S. Rappaport, “Wireless Communications: Principles and Practice,” 2nd Ed., Prentice Hall, 2002.
4. Yifei Yuan, Zhifeng Yuan, “5G New Radio Non-Orthogonal Multiple Access”, CRC Press, 2023.

24ICxxxT					Embedded Systems					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To introduce the foundational principles of embedded systems design and development.
- To acquire proficiency in programming microcontrollers and interfacing them with sensors, actuators, and other peripherals.
- To explore interfacing techniques for sensors, actuators, and other peripherals in embedded systems.
- To design and implement embedded systems solutions for real-world challenges through hands-on projects.

UNIT 1: EMBEDDED SYSTEMS – INTRODUCTION AND ARCHITECTURE	11 Hrs.
Introduction to embedded system (ES); Characteristics and functionalities; Classification of Embedded Systems; Microcontroller for Embedded Systems: microcontroller architecture, pin diagram and basic connections – clock, reset, brown-out, etc.; General register set; Special function registers; On-chip memories; General Purpose Inputs and Outputs (GPIOs)	
UNIT 2: ASSEMBLY PROGRAMMING FOR EMBEDDED SYSTEMS	**10 Hrs.**
Instruction Set and Assembly language programing: Data transfer, arithmetic, logic and branch instructions, addressing Modes; Basic programming concepts: Looping, branching, memory-access, time-delays, using assembly instructions; Stack and related operations; Subroutines and interrupts.	
UNIT 3: PERIPHERAL PROGRAMMING USING EMBEDDED C	**11 Hrs.**
Programming the on-chip modules/capabilities of the microcontroller using Embedded C: GPIOs, Timers/Counters, Waveform generation, EEPROM, ADC, Interrupt functionalities, Watchdog timer. Basics of parallel and serial communication; Programming the serial communication modules: USART, Serial Peripheral Interface (SPI), I2C Inter-integrated Circuit (I2C).	
UNIT 4: I/O INTERFACING IN EMBEDDED SYSTEM	**10 Hrs.**
Interfacing of I/O devices, LED and switches, 4x4 Keypad, 16x2 LCD, Motor drivers and opto-couplers, Stepper Motor, Servo Motor, DC Motors, Relays, Buzzers, Proximity sensor, temperature sensor, DAC, RTC. Interfacing wireless modules: Bluetooth, Wi-Fi, Zigbee.	

Total: 42 Hrs.

COURSE OUTCOMES

On completion of the course, the student will be able to:

CO1- Describe the basic components/functionalities, and architecture of microcontroller based embedded system. CO2- Write simple programs using assembly language, and summarize the instruction set architecture for microcontroller.

CO3- Apply the syntax and constructs of the C language, to write Embedded C programs.

CO4- Analyze the functionality of on-chip modules/capabilities of the microcontroller: timers/counters, interrupts, basic communication protocols, EEPROM, ADC, waveform generator.

CO5- Comprehend and demonstrate the working of external input-output devices (interfacing of sensors and actuators) to the microcontroller.

CO6- Develop simple embedded system, utilizing Embedded C programming for interfacing input-output devices with the microcontroller and/or the related on-chip modules.

TEXT/REFERENCE BOOKS

1. Muhammad Ali Mazidi, Sarmad Naimi, and Sepehr Naimi. "AVR Microcontroller and Embedded Systems: Using Assembly and C", (2015), Pearson India.
2. Raj Kamal, "Embedded Systems: Architecture, Programming, and Design", 2nd Edition, Tata McGraw-Hill.
3. Wayne Wolf, "Computers as Components: Principles of Embedded Computing System Design" – (second edition), Morgan Kaufmann
4. K.V. Shibu, "Introduction To Embedded Systems" - 2nd Edition, Tata McGraw-Hill

24ICxxxP					Embedded Systems Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES

- To impart knowledge of embedded system development.
- To familiarize with Microcontroller Boards and IDEs for Embedded System Development.
- To interface peripherals to Embedded Systems.
- To build a small scale Embedded System.

Laboratory Sessions would be based on following topics but not limited to:

1. Familiarization with IDE and trainer kits/boards.
2. Program for blinking LED pattern generation.
3. Program for interfacing multi-digit 7 segment display and implementing counter.
4. Program for interfacing toggle and push button switches, simple keypad, and matrix keypad.
5. Program for interfacing Buzzer, Relay, DC motor, Servo motor, Stepper motor.
6. Program for interfacing LCD and displaying text on it.
7. Program for interfacing various sensors and displaying quantity on LCD.
8. Program for interfacing speech and voice recognition modules and controlling peripherals.
9. Program for interfacing camera modules.
11. Program for interfacing RS 232 serial modules and file transfer using it.
12. Program for interfacing Ethernet module and transferring files using it.
13. Program for interfacing wireless modules: Zig bee, Bluetooth, GSM modem, GPS module, RFID etc.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1- Remember the basic components, and architecture of embedded systems.

CO2- Understand programs using assembly language and C programming.

CO3- Apply the concepts of interfacing for embedded systems.

CO4- Analyze different scenarios of embedded systems.

CO5- Evaluate the working of different embedded systems.

CO6- Create a small-scale embedded system for real world applications.

TEXT/REFERENCE BOOKS

1. Shibu K. V, Introduction to Embedded Systems, TMH.
2. Frank Vahid, Tony Givargis, Embedded system design: A unified Hardware/Software introduction, Wiley.
3. Rajkamal, Embedded System: Architecture, Programming and Design, TMH.
4. Wayne Wolf, Morgan, Computer as Components: Principles of Embedded Computing System Design, Kaufmann Publication.
5. Muhammad Ali Mazidi, 8051 Microcontroller and Embedded Systems: Using Assembly and C, Pearson education India.
6. Muhammad Ali Mazidi, AVR Microcontroller and Embedded Systems: Using Assembly And C, Pearson education India.
7. Muhammad Ali Mazidi, PIC Microcontroller and Embedded Systems: Using Assembly And C, Pearson education India.

XXXX					Digital CMOS VLSI Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	0	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the characteristics and concepts related to the design of digital CMOS VLSI circuits/gates.
- To explore various CMOS logic styles, and design CMOS VLSI circuits/gates at the transistor level and layout level.
- To analyze the performance/power of digital CMOS VLSI circuits/gates.

UNIT 1 INTRODUCTION TO DIGITAL CMOS VLSI	08 Hrs.
Trends in VLSI/Semiconductor industry, related to technology (Moore‟s) scaling; ITRS Roadmap; Overview of semiconductor devices inherent in the MOSFET: MOSFET I/V Characteristics, MOSFET C/V Characteristics, Second Order Effects. DC Transfer Characteristics: CMOS inverter DC Characteristics, Beta Ratio Effects, Noise Margins, Pass Transistor DC Characteristics	
UNIT 2: CIRCUIT CHARACTERIZATION AND PERFORMANCE ESTIMATION	12 Hrs.
RC Delay Model, Linear Delay Model, Logical Effort, Parasitic Delay, Delay in a Logic Gate, Delay in Multistage Logic Networks. Timing analysis delay models Power: Sources of power, Dynamic Power, Static Power, Energy-Delay optimization, Low-power design techniques.	
UNIT 3: DESIGN OF DIGITAL CMOS VLSI STANDARD AND COMPOUND GATES	14 Hrs.
Combinational Design: Circuit Families: Pass-transistor tree-based logic gates (and similar other logic styles – CPL, transmission gates, DPL, etc); Pseudo-NMOS logic; CVLS logic; Dynamic logic (domino, NP domino, Zipper); Hybrid logic style and examples (Full-Adder, MUX, XOR), Pitfalls of various families, Stick diagram. Sequential Design: Sequencing Static Circuits, Max-Delay Constraints, Min-Delay Constraints, Time Borrowing, Clock Skew, Circuit Design of Latches and Flip-Flops	
UNIT 4: Datapath and array subsystem design	8 Hrs.
Carry ripple adder, carry look ahead adder, Manchester carry chain adder, carry skip adder, carry select adder, shifters, multipliers, SRAM, DRAM, and other memories.	

Total: 42 Hours

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the pros and cons, and the trends related to scaling of the CMOS VLSI technology.

CO2 - Understand the characteristics, sub-parts, and regions of operation of the MOSFET, and use their I-V equations.

CO3 – Design of the combinational and sequential circuits using digital CMOS VLSI standard gates and compound gates, at the transistor –level.

CO4 - Estimate the performance/power of digital CMOS VLSI standard and compound gates.

CO5 - Analyze and compare various digital CMOS logic styles and gates/circuits.

CO6 - Create layouts of digital CMOS VLSI standard gates and compound gates, based on Lambda rules.

TEXT/REFERENCE BOOKS

1. Neil Weste, David Harris, Ayan Banerjee, “CMOS VLSI Design: A circuits and systems perspective”, 3rd Edition, Pearson.
2. Sung-Mo Kang, Yusuf Leblebici, “CMOS Digital Integrated Circuits”, 3rd Edition, Tata McGraw Hill.
3. Robert F. Pierret, “Semiconductor Device Fundamentals”, 1st Edition, Pearson.Prentice Hall.

XXXX					Digital CMOS VLSI Design Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	0	0	0	50	50	100

COURSE OBJECTIVES

- To learn how to create layout designs for CMOS circuits. This involves understanding the physical aspects of CMOS, like transistor placement, routing, and metal layers.
- To Use CAD (Computer-Aided Design) tools to simulate and verify the functionality of CMOS designs.
- To Gain proficiency in designing digital integrated circuits using CMOS technology. This includes understanding the basics of CMOS design, such as logic gates, flip-flops, and other digital building blocks.

Laboratory Sessions would be based on following topics:

1. To derive the CMOS INV gate noise margins using DC analysis.
2. To design layout of the CMOS INV and perform the RCX extraction and measure the delays.
3. To design layout of the CMOS NAND2 and perform the RCX extraction and measure the delays.
4. To design layout of the CMOS NOR2 and perform the RCX extraction and measure the delays.
5. To measure the dynamic and static power consumption of the CMOS NAND gate.
6. To understand the FO4 delay and measure the FO4 delay of the INV.
7. To apply the Logical effort technique to reduce the path delay.
8. To design layout of the Pass transistor AND2 and perform the RCX extraction and measure the delays.
9. To design layout of the DOMINO NAND2 and perform the RCX extraction and measure the delays.
10. To design layout of the Psuedo-NMOS NOR2 and perform the RCX extraction and measure the delays.

COURSE OUTCOMES

On completion of the course, students will be able to
CO1: Identify the pros and cons, and the trends related to VLSI Design.
CO2: Create layout designs for CMOS circuits and measure the performance metrics.
CO3: Understand the physical aspects of CMOS, including transistor placement, routing, and the use of metal layers.
CO4: Use CAD tools to simulate and verify the functionality of CMOS designs.
CO5: Perform static timing analysis to ensure that CMOS circuits meet their timing constraints.
CO6: Use techniques for minimizing power consumption in CMOS circuits. Understand the trade-offs between speed, power, and area.

TEXT/REFERENCE BOOKS

1. Neil Weste, David Harris, Ayan Banerjee, “CMOS VLSI design: A circuits and systems perspective”, 3rd Edition, Pearson.
2. Sung-Mo Kang, Yusuf Leblebici , “CMOS Digital Integrated Circuits”, 3rd Edition, Tata McGraw Hill.

Core Electives

Semester 7

24ECE***T					Advanced Processors and SoC					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To study the architecture of ARM series processors & DSP Processors.
- To understand the architecture and features of typical ARM7 microcontrollers and SoC devices.
- To understand memory and interconnect architecture for SoC.

UNIT 1: ARM family of processors	**12 Hrs.**
32/64-bit processors; introduction to ARM series processors and its versions: architectures, features, advantages & suitability in SoC design; ARM instruction set; ARM 7 based Microcontroller: features, architecture (block diagram and its description),Memory Map, GPIO, Pin Connect Block, Instruction set, programming in assembly language; Real-world interfacing with ARM7-basedMicrocontroller; overview of Cortex-A9 and Cortex-A15	
UNIT 2: Digital Signal Processors	**9 Hrs.**
Hardware architecture of DSP Processor: desirable features of DSP processors, types of architectures, internal architecture, features, system interface and instruction set of ADSP21xx, ADSP-21xx Development tools; TMS DSP processors; Overview of OMAP and AM57x series processors.	
UNIT 3: Memory Design for SoC	**10 Hrs.**
Introduction of SoC architecture; SoC external memory; Internal Memory: size, Scratchpads and Cache memory, Cache Organization, Cache data, Write Policies, Strategies for line replacement at miss time, Types of Cache, Split – I, and D –Caches, Multilevel Caches, SoC Memory System; Models of Simple Processor – memory interaction	
UNIT 4: Configuration and Interconnect Customization	**11 Hrs.**
An overview, Customizing Instruction Processor, Reconfiguration Technologies, Mapping design onto Reconfigurable devices, Reconfiguration – overhead analysis and trade-off analysis on reconfigurable Parallelism. Inter Connect Architectures: Basic Bus Architectures, Analytical Bus Models, Bus transactions and contention time; SoC Standard Buses (AMBA, Core connect).	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Define various advanced processors and their architectures.
CO2	:	Understand SoC Architectural features.
CO3	:	Illustrate memory and bus architecture for a SoC.
CO4	:	Analyze a computational task; characterize its computational requirements for a SoC
CO5	:	Select proper processor, memory and bus architecture for SoC configuration.
CO6	:	Develop an SoC framework for real world applications.

TEXT/REFERENCE BOOKS

1. Andrew Sloss, Dominic Symes, Chris Wright, "ARM System Developer"s Guide – Designing and Optimizing System Software", ELSEVIER
2. Phil Lapsley et al, "DSP Processor Fundamentals: Architectures and Features" Wiely India Pvt. Ltd
3. B. Venkataramani and M. Bhaskar, "Digital Signal Processors, Architecture, Programming and Applications" TMH.
4. Michael J. Flynn and Wayne Luk, "Computer System Design: System-on-Chip", Wiely India Pvt. Ltd.
5. Steve Furber, "ARM System on Chip Architecture", Addison Wesley Professional.
6. Ricardo Reis, "Design of System on a Chip: Devices and Components" Springer.

24ECE***T					Advance Communication Networks					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES:

- To learn 5G Technology advances and their benefits for next generation cellular networks
- To develop an intuitive understanding of various key wireless technologies unique to 5G networks.
- To know the propagation channel models and design characteristics of 5G.
- To learn 5G network architecture, enabling technologies, and supported services.
- To know the design and implementation issues of 5G networks.

UNIT-1: INTRODUCTION TO 5G BROADBAND WIRELESS COMMUNICATIONS. 11 Hrs.

Evaluation of mobile technologies 1G to 4G (LTE, LTEA, LTEA Pro), 5G Standardization, An Overview of 5G requirements, Regulations for 5G, Spectrum for 5G, Introduction to. LTE.

UNIT-2: 5G PHYSICAL LAYER AND APPLICATION 12 Hrs.

NR Physical layer overview, Propagation and Channel Modeling, Multi-carrier Waveforms, LTE/NR Interworking and Co-existence, NR in Unlicensed Spectrum, Industrial IoT and URLLC enhancements, 5G V2V Communication, 5G D2D Communication.

UNIT-3: 5G CHANNEL MODELING 10 Hrs.

V2V Channel Modeling in 5G communication; Geometric based Statistical MIMO Channel Modeling; 3D Scattering Channel Modeling for Microcell Environments; 3D Non-stationary Wideband UAV Channel.

UNIT-4: INTRODUCTION TO 6G 9 Hrs.

Moving towards 6G and Beyond cellular networks, 6G Key enabling technologies, CF-mMIMO, Introduction to mmWaves, AI in physical layer wireless communication 6G Use Cases and its key enabling technologies.

Max. Hrs.: 42

COURSE OUTCOMES:

After completion of this course, the student will be able to

CO1 - Remember the terminologies of cellular networks, their advances and benefits for next generation cellular networks.
CO2 - Understand the key RF, PHY, MAC layer and air interface changes required to support the requirements of 5G
CO3 - Apply various propagation channel models and design characteristics on 5G cellular architecture.
CO4 - Analyze different blocks of 5G network architecture and examine their functionalities.
CO5 - Evaluate the performance of Device-to-Device communication and millimeter wave communication in 5G.
CO6 - Understand the use cases for 6G Communications

TEXT/REFERENCE BOOKS:

1. Martin Sauter "From GSM From GSM to LTE–Advanced Pro and 5G: An Introduction to Mobile Networks and Mobile Broadband", Wiley-Blackwell.
2. Afif Osseiran, Jose.F.Monserrat, Patrick Marsch, "Fundamentals of 5G Mobile Networks" , Cambridge University Pres
3. Paulo Ramjee Prasad, Paulo Sergio Rufino Henrique, "6G The Road to the Future Wireless Technologies 2030" River Publishers.

XXXXXX					IC Technology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Mark
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES:

- To provide understanding of common trends and techniques in semiconductor manufacturing.
- To develop an in-depth understanding of the mechanism and physics of semiconductor manufacturing processes.
- To provide a foundation in semiconductor technology to address the challenges of advanced electronic devices.

UNIT I: INTRODUCTION, CRYSTAL STRUCTURE AND GROWTH, AND CLEANROOM 10 Hrs.

Moore‟s law, CMOS scaling, Overview of key semiconductor technologies, Crystal Structure: Bravais lattice, Miller‟s indices, Crystal planes; Crystal defects - point defects, Linear defects, and volume defects; Raw materials and purification, Crystal growth: Czochralski and Float-Zone processes, Silicon wafers, their specifications and measurements; Contamination Control: Cleanrooms - features, classifications and guidelines; Wafer Cleaning, Gettering.

UNIT II: LITHOGRAPHY AND OXIDATION PROCESSES 10 Hrs.

Photolithography: introduction and its application to IC manufacturing, Light sources, Photomask, Wafer exposure system: contact proximity and projection printing, Immersion lithography, Resolution and depth of focus, Photoresists: types, and issues; Process flow for lithography; EUV and electron-beam lithography. Oxidation: Dry vs wet oxidation, LOCOS, Thermal oxidation furnaces: Horizontal vs. vertical - their mechanism, Si/SiO2 interface, Oxide growth charges; growth kinetics, Deal-Grove model, Physical measurement: ellipsometry, measurement by CV method.

UNIT III: DIFFUSION, ION-IMPLANTATION AND THIN-FILM DEPOSITION 12 Hrs.

Doping process: doping profile of PN-junction, Trends in diffusion, Solid solubility, Fick‟s second law and its analytical solution, Gaussian solutions, Diffusion furnaces - horizontal vs. vertical; Ion implantation process: Dose, Energy and Angle considerations, Implantation damage and annealing - RTA, Channeling and shadowing effect, Diffusion vs. ion-implantation and their applications. Thin film deposition: requirements, characteristics and step coverage; PVD techniques: thermal and e-beam, Sputtering: types - DC and RF sputtering, process parameters and applications; CVD techniques and types: APCVD, LPCVD, PECVD, MOCVD; Atomic layer deposition (ALD) techniques: thermal and PEALD; Epitaxy.

UNIT IV: ETCHING AND BACK-END TECHNOLOGY 10 Hrs.

Etching: types - wet and dry etch, anisotropic and isotropic, selectivity, plasma etching and RIE; Wafer testing and sorting: Probe card setup, Electrical testing, Test data collection, Yield analysis, Defect mapping and diagnosis, Die sorting. Back-end technology: Interconnects in CMOS technology, Metallic interconnects, Copper interconnects – Advantages, Challenges posed, Interconnect fabrication process. Packaging assembly technology: Wafer thinning, Dicing, Die-attach, Wire-bond & Flip-chip process, Encapsulation, Laser marking, Solder ball attach, Singulation.

Max Hrs.: 42 Hrs.

COURSE OUTCOMES:

After completion of this course, the student will be able to

CO1: Understand the general trends and common techniques in semiconductor manufacturing.
CO2: Learn about cleanroom protocols, safety, gowning, and contamination control techniques.
CO3: To gain a comprehensive understanding of lithography, oxidation, diffusion, and ion implantation processes.
CO4: Analyze the fabrication techniques, trade-offs, and key considerations in developing a device.
CO5: Gain the understanding of front-end and back-end process altogether to apply this in an industry environment.
CO6**:** Build upon a foundation to pursue research and development of future semiconductor technologies.

TEXT/REFERENCE BOOKS:

1. J. D. Plummer, M. D. Deal & P. B. Griffin, Silicon VLSI Technology, Prentice-Hall, 2000.
2. C. Y. Chang and S. M. Sze –ULSI Technology, McGraw-Hill series, 1996.
3. R.C. Jaeger, Introduction to microelectronic fabrication, Prentice Hall, Second Edition, 2013.

24ECExxxT					Drones: Design, Theory and Applications					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Identify drone types and their respective use cases.
- Describe sensor, control, and communication system integration in drone design.
- Apply concepts to design drone systems and subsystems.
- Analyze the integration of sensors, actuators, and controls in drone design.

UNIT I: INTRODUCTION TO DRONES and ITS MECHATRONIC COMPONENTS	**08 Hrs.**
Overview of drone technology: history, evolution, and current trends, Types of drones: fixed-wing, rotary-wing (quadcopters, hexa and octacopters, etc.), hybrid. Components of a drone: frame, motors, propellers, electronic flight controllers, electronic speed controllers, sensors, communication systems, power source, etc. Drone Design Aspects, Regulations in drone operations. Safety measures and best practices for drone operation.	
UNIT II: DRONE DESIGN ASPECTS	**10 Hrs.**
Quadrotors, Key components of Autonomous flight, State estimation, Controls, Applications, Basic Mechanics, Dynamics and 1-D Linear Control, Design Considerations, Agility and Manoeuvrability, Component Selection, Effects of Size, Dynamical Systems and Rates of Convergence	
UNIT III: DRONE MODELLING & SIMULATION TOOLS	**13 Hrs.**
Transformations, Rotations, Euler Angles, Axis/Angle Representations For Rotations, Angular Velocity, Rigid-Body Displacements, Properties Of Functions, Symbolic Calculations in Matlab, Eigenvalues And Eigenvectors of Matrices, Quaternions, Matrix Derivative, Skew-Symmetric Matrices and the Hat Operator, Formulation, Newton-Euler Equations, Principal Axes And Principal Moments of Inertia, Quadrotor Equations Of Motion, Linear and Nonlinear State-Space Forms, Drone Modelling in Python and MATLAB/Simulink, ROS and its Libraries for Drones, Gazebo environment for Aerial robots.	
UNIT IV: AUTONOMOUS DRONE FLIGHT IN PRACTICAL APPLICATIONS	**11 Hrs.**
2-D Quadrotor Control, 3-D Quadrotor Control, Time, Motion, and Trajectories, Time, Motion, and Trajectories, Motion Planning for Quadrotors, Minimum Velocity Trajectories from the Euler-Lagrange Equations, Solving for Coefficients of Minimum Jerk Trajectories, Linearization of Quadrotor Equations of Motion, Multi-drone Applications: Inspection, Surveillance, Monitoring, Transportation, Agriculture. Drone with Mobile Robot operations.	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to:

CO1	:	Identify various types of drone and classify their use cases.
CO2	:	Describe the integration of sensors, control and communication systems in drone design.
CO3	:	Apply concepts in modelling and simulation of drone systems and subsystems
CO4	:	Analyse the integration of sensors, actuators communication systems in drone design.
CO5	:	Evaluate the suitability of different control algorithms and navigation systems for specific drone missions.
CO6	:	Design innovative solutions to real-world challenges in drone design, operation, and applications.

TEXT/REFERENCE BOOKS

1. Randal W. Beard and Timothy W. McLain " Small Unmanned Aircraft: Theory and Practice", Princeton University Press, 2012.
2. Kenzo Nonami "Autonomous Flying Robots: Unmanned Aerial Vehicles and Micro Aerial Vehicles" , Springer, 2010
3. Quan Quan, Xunhua Dai, Shuai Wang "Multicopter Design and Control Practice - A Series Experiments based on MATLAB and Pixhawk" , Springer, 2020

<Course Code>					Radar and Navigation systems					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES-

- Understand the basic principles of different types of RADAR
- Gaining in-depth knowledge of radar transmitter and receivers
- Exhibit RADAR navigation techniques

UNIT I: 1RADAR FUNDAMENTALS	**10 Hrs.**
Introduction to Radar: Basic Radar – The Origins of Radar, radar system (block diagrams), Radar range Equation, Applications of Radar. Radar types: MTI, Doppler and Pulse, PRF, Delay, Line Cancellers, Staggered Pulse Repetition Frequencies, Doppler Filter Banks, Digital MTI Processing, Moving Target Detector, Limitations to MTI Performance, Pulse Doppler Radar. Tracking with Radar-Monopulse Tracking, Conical Scan and Sequential Lobing, Limitations to Tracking Accuracy, Low-Angle Tracking - Tracking in Range, Comparison of Trackers, Automatic Tracking with Surveillance Radars	
UNIT II: RADAR RECEIVERS	**10 Hrs.**
Introduction, Superheterodyne Receiver, Receiver noise Figure, Duplexers and Receiver Protectors, Radar Displays. Matched Filter Receiver, Detection Criteria, Detectors, Automatic Detector, Integrators, Constant-False-Alarm Rate Receivers, The Radar operator, Signal Management, Propagation Radar Waves, Atmospheric Refraction, Standard propagation, Nonstandard Propagation, The Radar Antenna, Reflector Antennas, Electronically Steered Phased Array Antennas, Phase Shifters, Frequency-Scan Arrays.	
UNIT III: RADAR TRANSMITTERS AND SIGNAL DETECTION	**10 Hrs.**
Introduction –Linear Beam Power Tubes - Solid State RF Power Sources - Magnetron -Crossed Field Amplifiers - Other RF Power Sources – Other aspects of Radar Transmitter. Detection of Signals in Noise –Detection of Signals in Noise, Receiver Noise and the Signal-to-Noise Ratio, Probability Density Functions, Probabilities of Detection and False Alarm, Integration of Radar Pulses, Radar Cross Section of Targets, Radar cross Section fluctuations, Transmitter Power	
UNIT IV: NAVIGATION	**12 Hrs.**
Introduction, Four methods of Navigation, Radio Direction Finding, The Loop Antenna, Loop Input Circuits, An Aural Null Direction Finder, The Goniometer, Errors in Direction Finding, Adcock Direction Finders, Direction Finding at Very High Frequencies, Automatic Direction Finders, The Commutated Aerial Direction Finder, Range and Accuracy of Direction Finders, Radio Ranges, Doppler Navigation, component, Beam Configurations, Track Stabilization, introduction to Satellite Navigation System, Global Positioning System(GPS).Instrument landing system, microwave landing system	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES-

On completion of the course, student will be able to:

CO1	:	Remember the basic fundamentals of communication receiver.
CO2	:	Understand the different types of RADAR and their working.
CO3	:	Apply the basic concept to understand the detection of signal.
CO4	:	Analyze the working of RADAR Transmitter, receiver and RADAR
CO5	:	Evaluate various RADAR and Navigation systems and components.
CO6	:	Design components for RADAR and Navigation system.

TEXT/REFERENCE BOOKS -

1. Merrill I. Skolnik ," Introduction to Radar Systems", Tata McGraw-Hill.
2. Peyton Z. Peebles:, "Radar Principles", John Wiley Publication.
3. J.C Toomay, " Principles of Radar", PHI.

<Course Code>					Internet of Things					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Mark
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To impart knowledge about components of IoT, architecture, and platforms of IoT.
- To apprise students with basic knowledge of IoT that paves a platform to understand physical, logical design, and business models.
- To teach a student how to analyse requirements of various industrial systems for cost-effective design of IoT applications on different IoT platforms.
- To explain the students how to design IoT applications in different domains and real time scenarios

UNIT 1 INTRODUCTION	**11 Hrs.**
Definition & Characteristics of IoT, Technology evolution, IoT Architectures, Physical and Logical Design of IoT, IoT Functional Blocks, IoT Communication Models- Publish-subscribe, Push-Pull, exclusive-pair models, Resource management, IoT Security, Current challenges and issues.	
UNIT 2 IoT DEVICES AND TECNOLOGIES	**12 Hrs.**
Introduction to various types of sensors, Working principles of sensors, Actuators, Control Units-Microcontrollers and Microprocessors, I/O interfaces, Communication Technologies for IoT, RFID, IoT Enabling Technologies – AI, Bigdata Analytics, Cloud Computing, and Embedded Systems.	
UNIT 3 INDUSTRIAL IoT	**9 Hrs.**
Introduction to Industrial IoT, Role of Industrial Internet of Things (IIoT), Difference between IoT and IIoT, Business Models, IoT- based real-time control systems and Human-Machine Interface, Supervisory Control & Data Acquisition (SCADA), Case studies on Smart Factories with IIoT.	
UNIT 4 IoT SYSTEMS AND APPLICATIONS	**10 Hrs.**
Applied Internet of Things, Smart Cities, Smart Agriculture, Intelligent Transportation Systems, IoT Applications for Smart Health Care, Home Automation, Environment Monitoring, and Surveillance, Case Studies Illustrating IoT Design.	

Max. 42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to

CO1	:	Remember the Internet of Things and its technological aspects.
CO2	:	Comprehend advanced IoT devices and technologies for different IoT applications.
CO3	:	Apply the knowledge of control units and communication technologies in designing IoT systems.
CO4	:	Analyse the Industrial systems with IoT components.
CO5	:	Evaluate efficiency trade-offs among alternative models for an efficient IoT application design
CO6	:	Design different IoT systems for various real-life problems and application domains

TEXT/REFERENCE BOOKS

1. Pethuru Raj and Anupama C. Raman. The Internet of Things: Enabling Technologies, Platforms, and Use Cases. CRC Press.
2. Adrian McEwen. Designing the Internet of Things. Wiley publications.
3. Vijay Madisetti, Arshdeep Bahga. İnternet of Things: A Hands-on Approach. University Press.
4. Raj Kamal. Internet of Things: Architecture and Design. McGraw Hill

<Course Code>					Mixed Signal VLSI Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To learn the fundamentals of mixed-signal circuit design.
- To study of data converters (adc/dac) for interfacing of analog and digital systems.
- To know the various methods for the co-existence of analog and digital circuits on a single chip.

UNIT I: INTRODUCTION TO MIXED-SIGNAL	10 Hrs.
Definition of Mixed-signal ICs, Challenges, Applications, Basic to data conversion and processing, Sampling, Switched capacitors circuits: MOSFET as switches, Diode as Switch, Switched Capacitor Integrator, MOS sample-hold circuits, Current mirror, Voltage-Controlled Oscillators (VCO), Phase Locked Loop (PLL), non- ideal effect in PLL, Jitter and phase	
UNIT II: ANALOG TO DIGITAL CONVERTERS	**12 Hrs.**
Performance Metric, Flash Architectures, Two-Step Architectures, Interpolative and Folding Architectures, Pipelined Architectures, Successive Approximation Architectures, Interleaved Architectures, Over sampling ADC – Noise shaping, Sigma- Delta modulator.	
UNIT III: DIGITAL TO ANALOG CONVERTERS	**10 Hrs.**
Performance Metrics, Reference Multiplication and Division: Voltage Division, Current Division, Charge Division, Switching and logical Functions in DACs, Switching Functions in Resistor-Ladder DACs, Switching Functions in Current-Steering DACs, Switching Functions in Capacitor DACs, Binary-to-Thermometer Code Conversion, **Architectures:** Resistor-Ladder DAC Architectures, Ladder Architecture with Switched Sub-divider, Intermeshed Ladder Architectures, Current-Steering Architectures, R-2R-Network Based Architectures, Segmented Architectures	
UNIT IV: NOISE AND TESTING OF ICs	**10 Hrs.**
Noise: Types of noise: Shot, Thermal, Flicker (1/f), Burst, and avalanche noise, Noise Models of IC components: Diode, BJT, MOS transistor, Resistor, capacitors, and inductors, Noise power trade-off, Noise Bandwidth, Noise Figure, Noise Temperature, Total harmonic distortion. **Testing:** General Consideration, Sampling circuits, D/A converters, A/D convertors: static and dynamic testing. **Project**	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Define the importance of mixed-signal circuits in ic design.
CO2	:	Understand the principles and mathematical concepts of different mixed-signal circuit architectures.
CO3	:	Apply the appropriate amplifier circuit to amplify analog and discrete signals.
CO4	:	Analyze and compare the performance metrics of data converters.
CO5	:	Evaluate the various a/d and d/a converters for a given data acquisition system.
CO6	:	Design the mixed-signal circuit ic for real-life applications.

TEXT/REFERENCE BOOKS

1. Behzad razavi, "design of analog cmos integrated circuits," mcgraw-hill international edition 2016.
2. Behzad razavi, "principles of data conversion system design," wiley-ieee press, 1995
3. R. Jacob baker, "cmos mixed-signal circuit design," wiley india, ieee press, and reprint 2008.
4. Tony chan carusone, david a. Johns, kenneth w. Martin, "analog integrated circuit design," wiley, 2nd edition.

24ECxxxT					Deep Learning and Applications					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Introduce concepts of artificial neural networks.
- Understand concepts and applications of deep learning using artificial neural networks.
- Understand concepts and requirements of adaptive learning techniques.

UNIT 1: Introduction To Artificial Neural Networks	**10 Hrs.**
Introduction to Neural networks, Perceptron, McCulloch Pitts Neuron, Models of Neuron, Artificial Neural Network (ANN) Architectures, Classification of ANNs, ANN learning, analysis and applications, Single Layer & Multi-Layer Perceptron (MLP) Network, Optimization of neural networks (GS, SGD, Adagrad, Adadelta, RMSprop, Adam, NAG). Bias vs. variance trade-off, Regularization.	
UNIT 2: Convolutional Neural Networks	**11 Hrs.**
Introduction to Deep Learning (DL) Techniques, Convolutional Neural Networks (CNN), Basic architecture, Convolutional Layers, Pooling Layers, Activation functions, Handling vanishing gradient problem, Dropout, Batch Normalization; Different CNN Models: LeNet, AlexNet, VGGNet, GoogleNet, ResNet, DenseNet, Training of Deep neural Networks, Hyper parameter tuning, transfer learning, Dataset augmentation.	
UNIT 3: Deep Learning Neural Networks for Time Series Data	**11 Hrs.**
Feedforward and Recurrent Neural Networks (RNN), Back propagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, Bidirectional RNNs, Deep RNN, Long Short-Term Memory Networks (LSTMs), Bidirectional LSTMs, Gated Recurrent Unit (GRU), Application of Deep Learning Networks in Various Forecasting Problems.	
UNIT 4: Neural Networks for Special Applications	**10 Hrs.**
Introduction to Encoder Decoder Models, Variational Autoencoders, GAN and Different Type of GAN"s, Implementation on GAN"s, Siamese network, Attention and Transformers, Types and Applications.	

Total: 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Define the basic concepts and terminology associated with neural networks, including neurons, activation functions, and network architectures.

CO2: Explain the fundamental principles underlying neural network operation, including forward and backward propagation, gradient descent, and optimization techniques.

CO3: Implement and train simple feedforward neural networks for tasks such as classification and apply appropriate preprocessing techniques and hyperparameter tuning to improve the performance of neural networks on real-world datasets.

CO4: Evaluate the strengths and weaknesses of different neural network architectures and activation functions for specific applications.

CO5: Evaluate the performance of advanced neural network architectures (e.g., convolutional neural networks, recurrent neural networks) on challenging tasks like image recognition, natural language processing, and time series prediction.

CO6: Design and implement novel neural network architectures or modifications to existing architectures to address specific challenges or improve performance on specialized tasks.

TEXT/REFERENCE BOOKS

1. Simon Haykin, Neural Networks and Learning Machines, Third Edition, Paper Back, Pearson India.

2. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press.
3. Josh Patterson and Adam Gibson, Deep Learning A Practitioner‟s Approach, O‟Reilly Media, Inc.
4. Antonio Gulli, Sujit Pal, Deep Learning with Keras, Packt Publishing.
5. Christopher Bishop, “Deep Learning: Foundations and Concepts”, Springer.